

AYMANE AIT DADS

AI Research Intern - Agentic AI | LLM Fine-Tuning · Agentic AI · Transformer Architecture

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PROFESSIONAL SUMMARY

Achieved top 10% public leaderboard (0.80/1.0) in the NVIDIA Nemotron Reasoning Challenge (\$106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE with LoRA and conducting 20+ single-variable ablation studies on chain-of-thought reasoning tasks. Ranked 1st in EURECOM class for AI-generated image detection (0.791 AUC) using CLIP ViT-L/14 fine-tuning and submitted to the NTIRE 2026 @ CVPR international benchmark, demonstrating end-to-end research methodology from prototype to academic benchmark. Technical expertise spans LLM fine-tuning (LoRA, QLoRA, SFTTrainer), agentic AI orchestration (Claude API, tool-calling, MCP server, RAG, LangChain), and deep learning frameworks including PyTorch, Hugging Face Transformers, and PEFT. Available from Summer 2026 for an internship in Europe; no work authorization required (EU-based student at EURECOM, Sophia Antipolis, France).

CORE COMPETENCIES

Agentic AI | LLM Fine-Tuning | Transformer Architecture | Chain-of-Thought Reasoning | Ablation Studies | Prompt Engineering | RAG | Mobile AI Research

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a Random Forest + K-Means pipeline on **100,000+ fiber-optic network records** to classify quality across thousands of nodes, directly informing network optimization decisions.
- Designed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for maintenance planning.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reproducible Scikit-learn pipelines built in Python and Pandas.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, bridging the gap between data science results and business decision-making.

PROJECTS

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard score of 0.80/1.0** (\$106K+ prize pool) fine-tuning Nemotron-3-Nano-30B, a hybrid 30B Mamba-Transformer MoE (3.5B active params via MoE routing), with a LoRA adapter (r=32, ~888M trainable parameters) on 7,828 verified chain-of-thought examples across 6 logic puzzle types evaluated by exact match.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order - including monkey-patched VRAM allocator, stubbed Mamba/Cutlass imports, disabled fast-path CUDA kernels, mocked Triton ptxas, and pure-PyTorch RMSNorm fallback - and ran 20+ single-variable ablations confirming completion-only loss (+0.01 AUC), packing=False (+0.16), and all-linear 888M LoRA (+0.22 vs. 12-layer baseline) as critical design choices.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · 1st in Class

- Ranked **1st in EURECOM class private Kaggle leaderboard (0.791 AUC)** fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head and dual learning rates, training on 250K images spanning 25+ generator types with FP16 mixed precision on a T4 GPU.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench international benchmark**; validated that 1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC) via ablation, and applied a 3-model ensemble with 10-view TTA to maximize out-of-distribution generalization.

PyTorch · OpenCLIP · Hugging Face Transformers · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Image Security, Cloud Computing, Statistics, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & Agentic AI: LoRA / QLoRA | SFTTrainer | Mamba-Transformer MoE | vLLM | Claude API | Tool-Calling | RAG | LangChain | MCP Server | Prompt Engineering

Deep Learning & Computer Vision: PyTorch | Hugging Face Transformers | PEFT | OpenCLIP | ViT | CNN | DenseNet | TensorFlow | Scikit-learn

MLOps & Data Science: Python | SQL | Bash | Git | CUDA Debugging | Triton | FP16/BF16 Mixed Precision | Pandas | NumPy | Power BI